

BS-PH101

Physics-I

Combined Module Notes

Modules 1 – 5

Semester I/II • ECE • 4 Credits

CGEC | B.Tech ECE (2023–27)

Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Exam-Ready Notes

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Module 1

Constraints and Friction

1. Constraints and Friction

1.1 Constraints and Degrees of Freedom

Definition: Constraint and Degrees of Freedom

A **constraint** is any physical restriction on the position or motion of a mechanical system. The number of independent coordinates required to completely specify the configuration of a system is called its **Degrees of Freedom (DOF)**.

If a system consists of N particles in 3D space subject to k independent constraint equations, the number of degrees of freedom is given by:

$$f = 3N - k \quad (1)$$

Classification of Constraints:

Constraint Type	Condition / Mathematical Form	Physical Example
Holonomic	Expressible as $f(r_1, r_2, \dots, r_N, t) = 0$ (integrable algebraic relations).	Rigid body ($ \vec{r}_i - \vec{r}_j = c$), simple pendulum ($r = L$).
Non-Holonomic	Expressible as inequalities or non-integrable differential relations ($f(\vec{r}, \dot{\vec{r}}, t) = 0$).	Rolling without slipping on a rough plane, gas molecules in a container ($r \leq R$).
Scleronomic	Constraint equations do not explicitly depend on time t ($\partial f / \partial t = 0$).	Bead on a fixed wire ring, rigid dumbbell.
Rheonomic	Constraint equations explicitly depend on time t ($\partial f / \partial t \neq 0$).	Bead on a wire ring rotating with constant angular velocity ω .

1.2 Friction and Laws of Friction

Friction is the resistive contact force that opposes the relative motion or tendency of relative motion between two surfaces in contact.

Key Laws and Relations of Friction

- **Static Friction (f_s):** Self-adjusting force. Maximum value is limiting friction:

$$f_{s,\max} = \mu_s N \quad (2)$$

- **Kinetic Friction (f_k):** Constant resistive force during motion:

$$f_k = \mu_k N \quad (\text{where } \mu_k < \mu_s) \quad (3)$$

- **Angle of Friction (λ):** Angle between total contact reaction $\vec{R}$ and normal reaction $\vec{N}$:

$$\tan \lambda = \frac{f_{s,\max}}{N} = \mu_s \implies \lambda = \tan^{-1}(\mu_s) \quad (4)$$

- **Angle of Repose (θ_r):** Maximum angle of inclination of a rough plane at which a body remains at rest without sliding:

$$mg \sin \theta_r = \mu_s mg \cos \theta_r \implies \tan \theta_r = \mu_s \implies \theta_r = \lambda \quad (5)$$

Solved Problem 1: Block on an Inclined Plane with Friction

Problem: A block of mass $m = 5$ kg rests on an inclined plane of angle $\theta = 30^\circ$. The coefficient of static friction is $\mu_s = 0.4$. Determine whether the block slides down, and find the frictional force.

Solution:

1. Component of gravity along the incline: $F_{\text{down}} = mg \sin 30^\circ = 5 \times 9.8 \times 0.5 = 24.5$ N.
2. Normal reaction: $N = mg \cos 30^\circ = 5 \times 9.8 \times 0.866 = 42.435$ N.
3. Maximum limiting static friction: $f_{s,\max} = \mu_s N = 0.4 \times 42.435 = 16.97$ N.
4. Since $F_{\text{down}}(24.5 \text{ N}) > f_{s,\max}(16.97 \text{ N})$, the block **will slide down**. The friction force acting up the plane during motion is kinetic friction $f_k = \mu_k N$.

Solved Problem 2: Degrees of Freedom of Connected Bodies

Problem: Find the degrees of freedom for: (a) A double pendulum moving in a plane, (b) Two particles connected by a rigid rod free to move in 3D space.

Solution:

1. **Double Pendulum in 2D:** $N = 2$ particles, constrained to plane ($z_1 = 0, z_2 = 0$), fixed length $r_1 = L_1$ and $|\vec{r}_2 - \vec{r}_1| = L_2$. Total independent angles θ_1, θ_2 . Hence DOF = 2.
2. **Rigid Dumbbell in 3D:** $N = 2$ particles in 3D (6 coordinates) subject to $k = 1$ constraint ($|\vec{r}_1 - \vec{r}_2| = d$). Hence DOF = $3(2) - 1 = 5$ (3 translational, 2 rotational).

2. Vector Calculus and Potential Energy Function

2.1 Gradient, Divergence, and Curl

Definition: Gradient, Divergence, and Curl

- **Gradient** (∇V): Vector representing the rate and direction of maximum spatial change of a scalar field $V(x, y, z)$.
- **Divergence** ($\nabla \cdot \vec{A}$): Scalar representing the net outward flux of a vector field per unit volume.
- **Curl** ($\nabla \times \vec{A}$): Vector representing the circulation or rotational density of a vector field.

Mathematical Definitions in Cartesian Coordinates

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \quad (6)$$

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (7)$$

$$\text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (8)$$

2.2 Potential Energy Function $F = -\text{grad } V$

For a conservative force field $\vec{F}$, the work done along a closed path is zero ($\oint \vec{F} \cdot d\vec{r} = 0$). This implies $\vec{F}$ can be expressed as the negative gradient of a scalar potential energy function $V(\vec{r})$:

$$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}\right) \quad (9)$$

Physical Meaning of Negative Sign: The force acts in the direction of *decreasing* potential energy (towards stable equilibrium).

Equipotential Surfaces: A surface on which the potential energy $V(\vec{r}) = \text{constant}$.

- Work done in moving a mass along an equipotential surface is zero: $dW = \vec{F} \cdot d\vec{r} = -dV = 0$.
- The force vector $\vec{F}$ is everywhere perpendicular to the equipotential surface passing through that point.

Solved Problem 3: Deriving Force from Potential Energy

Problem: Given potential energy $V(x, y, z) = 3x^2y - y^3z^2$, calculate force $\vec{F}$ at point $(1, -1, 2)$.

Solution:

1. Calculate partial derivatives: $\frac{\partial V}{\partial x} = 6xy$, $\frac{\partial V}{\partial y} = 3x^2 - 3y^2z^2$, $\frac{\partial V}{\partial z} = -2y^3z$.
2. Evaluate at $(1, -1, 2)$: $\frac{\partial V}{\partial x} = 6(1)(-1) = -6$, $\frac{\partial V}{\partial y} = 3(1)^2 - 3(-1)^2(2)^2 = 3 - 12 = -9$, $\frac{\partial V}{\partial z} = -2(-1)^3(2) = 4$.
3. Substitute into $\vec{F} = -\nabla V$:

$$\vec{F} = -(-6\hat{i} - 9\hat{j} + 4\hat{k}) = 6\hat{i} + 9\hat{j} - 4\hat{k} \text{ N} \quad (10)$$

3. Conservative Forces and Conservation Laws

3.1 Conservative vs. Non-Conservative Forces

Comparative Breakdown:

Property	Conservative Force	Non-Conservative Force
Work Dependence	Work done depends only on initial and final positions.	Work done depends on the actual path taken.
Closed Path Work	$\oint \vec{F} \cdot d\vec{r} = 0$.	$\oint \vec{F} \cdot d\vec{r} \neq 0$.
Curl Condition	$\nabla \times \vec{F} = 0$ (Irrotational).	$\nabla \times \vec{F} \neq 0$.
Potential Energy	Expressible as $\vec{F} = -\nabla V$.	Potential energy function cannot be defined.
Mechanical Energy	Total mechanical energy ($E = K + V$) is conserved.	Mechanical energy is dissipated as heat/sound.
Examples	Gravitational force, Electrostatic force, Spring force.	Friction, Viscosity, Air resistance.

3.2 Fundamental Conservation Laws

Conservation Theorems

1. **Conservation of Linear Momentum:** If net external force $\vec{F}_{\text{ext}} = 0$, then:

$$\frac{d\vec{P}}{dt} = 0 \implies \vec{P} = \sum m_i \vec{v}_i = \text{constant} \quad (11)$$

2. **Conservation of Angular Momentum:** If net external torque $\vec{\tau}_{\text{ext}} = \vec{r} \times \vec{F}_{\text{ext}} = 0$, then:

$$\frac{d\vec{L}}{dt} = 0 \implies \vec{L} = \vec{r} \times \vec{P} = I\vec{\omega} = \text{constant} \quad (12)$$

3. **Conservation of Energy:** In an isolated system with conservative forces:

$$E_{\text{total}} = K + V = \frac{1}{2}mv^2 + V(\vec{r}) = \text{constant} \quad (13)$$

4. Non-Inertial Frames of Reference

4.1 Inertial vs. Non-Inertial Frames

Definition: Frames of Reference

- **Inertial Frame:** A frame of reference moving with zero acceleration (at rest or constant velocity) in which Newton's laws of motion hold strictly.
- **Non-Inertial Frame:** An accelerating or rotating frame of reference in which Newton's laws do not hold directly unless fictitious / pseudo forces are introduced.

4.2 Rotating Frames: Centrifugal and Coriolis Forces

Consider a frame S' rotating with constant angular velocity $\vec{\omega}$ relative to a stationary inertial frame S .

Transformation Equation for Acceleration

The transformation relation between acceleration in fixed frame $\vec{a}_S$ and rotating frame $\vec{a}_{S'}$ is:

$$\vec{a}_S = \vec{a}_{S'} + 2(\vec{\omega} \times \vec{v}') + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (14)$$

Multiplying by mass m , the effective force in the rotating frame is:

$$\vec{F}' = m\vec{a}_{S'} = \vec{F}_{\text{real}} - \underbrace{2m(\vec{\omega} \times \vec{v}')}_{\text{Coriolis Force } \vec{F}_{\text{cor}}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centrifugal Force } \vec{F}_{\text{cent}}} \quad (15)$$

Properties of Rotating Fictitious Forces:

Force	Vector Expression	Key Characteristics
Centrifugal Force	$\vec{F}_{\text{cent}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}) = m\omega^2 r_{\perp} \hat{r}$	Directed radially outward from axis of rotation. Present even if particle is stationary in rotating frame ($\vec{v}' = 0$).
Coriolis Force	$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$	Perpendicular to both velocity $\vec{v}'$ and angular velocity $\vec{\omega}$. Acts only when particle is in motion ($\vec{v}' \neq 0$). Does zero work ($W = 0$).

Physical Effects of Coriolis Force on Earth:

- **Deflection of Moving Objects:** Moving bodies in Northern Hemisphere are deflected to their *right*, while in Southern Hemisphere to their *left* (causes cyclone rotation direction).
- **Eastward Deflection of Freely Falling Body:** A body dropped from height h at latitude λ experiences an eastward deflection $\Delta x = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}}$.
- **Foucault Pendulum:** Demonstrates Earth's rotation directly via precession of its oscillation plane at angular rate $\Omega = \omega \sin \lambda$.

Solved Problem 4: Eastward Deflection Calculation

Problem: Calculate the eastward deflection due to Coriolis force for a body dropped from a height $h = 100$ m at Kolkata (latitude $\lambda = 22.5^\circ\text{N}$). (Take $\omega = 7.29 \times 10^{-5}$ rad/s, $g = 9.8$ m/s²).

Solution:

1. Formula for eastward deflection:

$$d = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}} \quad (16)$$

2. Calculate time of fall: $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{200}{9.8}} = 4.518$ s.

3. Substitute values: $d = \frac{1}{3}(7.29 \times 10^{-5})(9.8) \cos(22.5^\circ)(4.518)^3 \cos 22.5^\circ = 0.9239$, $(4.518)^3 = 92.23$. $d = 0.333 \times 7.29 \times 10^{-5} \times 9.8 \times 0.9239 \times 92.23 = 0.0195$ m = **1.95** cm.

5. Harmonic Oscillations

5.1 Damped Harmonic Motion

When an oscillator experiences a viscous damping force proportional to velocity ($F_{\text{damping}} = -b\dot{x}$), the equation of motion is:

$$m\ddot{x} + b\dot{x} + kx = 0 \implies \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0 \quad (17)$$

where $\gamma = \frac{b}{2m}$ is the damping coefficient and $\omega_0 = \sqrt{\frac{k}{m}}$ is the natural frequency.

Assuming solution $x(t) = Ce^{rt}$, the roots of characteristic equation are $r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$.

Three Cases of Damped Motion

1. **Overdamped** ($\gamma > \omega_0$): Non-oscillatory exponential return to equilibrium.

$$x(t) = e^{-\gamma t} \left(Ae^{\sqrt{\gamma^2 - \omega_0^2} t} + Be^{-\sqrt{\gamma^2 - \omega_0^2} t} \right) \quad (18)$$

2. **Critically Damped** ($\gamma = \omega_0$): Fastest non-oscillatory return to equilibrium without overshoot. Used in shock absorbers and deadbeat galvanometers.

$$x(t) = (A + Bt)e^{-\gamma t} \quad (19)$$

3. **Underdamped** ($\gamma < \omega_0$): Oscillatory motion with exponentially decaying amplitude.

$$x(t) = A_0 e^{-\gamma t} \cos(\omega_d t + \phi), \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \gamma^2} \quad (20)$$

Damping Parameters: Logarithmic Decrement and Quality Factor

- **Relaxation Time (τ):** Time in which energy drops to $1/e$ of initial value: $\tau = \frac{1}{2\gamma} = \frac{m}{b}$.
- **Logarithmic Decrement (δ):** Natural logarithm of ratio of successive amplitudes one period T_d apart:

$$\delta = \ln \left(\frac{A_n}{A_{n+1}} \right) = \gamma T_d = \frac{b T_d}{2m} \quad (21)$$

- **Quality Factor (Q):** Dimensionless measure of sharpness of oscillation energy retention:

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma} = \frac{\omega_0 \tau}{1} = \frac{\pi}{\delta} \quad (22)$$

5.2 Forced Oscillations and Resonance

Under a periodic external driving force $F(t) = F_0 \cos(\omega t)$, the differential equation is:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad \left(\text{where } f_0 = \frac{F_0}{m} \right) \quad (23)$$

In steady state, the response oscillates at driving frequency ω :

$$x(t) = x_0 \cos(\omega t - \phi) \quad (24)$$

where steady-state amplitude x_0 and phase lag ϕ are given by:

$$x_0(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2 \omega^2}}, \quad \tan \phi = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \quad (25)$$

Resonance Conditions

- **Amplitude Resonance Frequency (ω_r):** Frequency at which displacement amplitude x_0 is maximum:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}, \quad x_{0,\max} = \frac{f_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{f_0}{2\gamma\omega_0} \quad (\text{for small } \gamma) \quad (26)$$

- **Velocity Resonance Frequency (ω_v):** Frequency at which velocity amplitude $v_0 = \omega x_0$ is maximum:

$$\omega_v = \omega_0 \quad (\text{Natural undamped frequency}) \quad (27)$$

- **Phase at Resonance:** At velocity resonance ($\omega = \omega_0$), $\tan \phi = \infty \implies \phi = 90^\circ$ (force and velocity are in phase).

6. Rigid Body Dynamics and Moment of Inertia

6.1 Rigid Body Motion and Angular Velocity Vector

A rigid body is a system of particles in which inter-particle distances remain constant under external forces ($|\vec{r}_i - \vec{r}_j| = \text{constant}$).

General motion of a rigid body is a combination of translation of center of mass (CM) and rotation about CM. The angular velocity vector $\vec{\omega}$ is a vector directed along the instantaneous axis of rotation according to the right-hand rule.

Linear velocity of any particle i at position $\vec{r}_i$ relative to an axis is:

$$\vec{v}_i = \vec{\omega} \times \vec{r}_i \quad (28)$$

6.2 Moment of Inertia and Parallel/Perpendicular Axes Theorems

Definition: Moment of Inertia

Moment of Inertia (I) measures the rotational inertia of a body about a given axis. For a continuous mass distribution:

$$I = \sum m_i r_i^2 = \int r^2 dm = mk^2 \quad (29)$$

where k is the **radius of gyration**.

Fundamental Theorems of Moment of Inertia

1. **Parallel Axis Theorem:** MI about any axis is equal to MI about a parallel axis through CM plus Mh^2 :

$$I = I_{\text{cm}} + Mh^2 \quad (h = \text{perpendicular distance between parallel axes}) \quad (30)$$

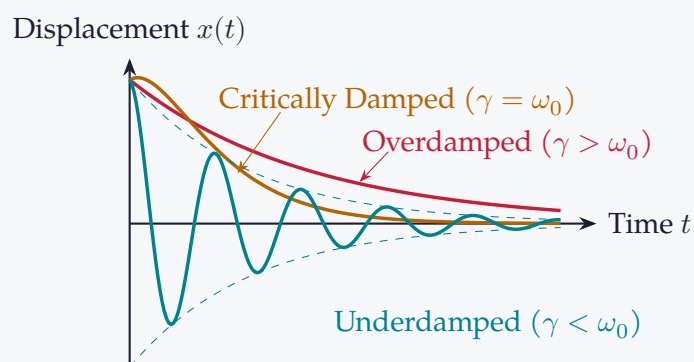
2. **Perpendicular Axis Theorem:** For a planar lamina in xy -plane, MI about z -axis perpendicular to lamina is:

$$I_z = I_x + I_y \quad (31)$$

Standard Moments of Inertia Summary:

Body	Axis of Rotation	Moment of Inertia (I)
Thin Uniform Rod (Length L)	Perpendicular through center.	$I = \frac{1}{12}ML^2$
Thin Uniform Rod (Length L)	Perpendicular through one end.	$I = \frac{1}{3}ML^2$
Circular Ring (Radius R)	Central axis perpendicular to plane.	$I = MR^2$
Circular Disc (Radius R)	Central axis perpendicular to plane.	$I = \frac{1}{2}MR^2$
Solid Sphere (Radius R)	Any diameter axis.	$I = \frac{2}{5}MR^2$
Spherical Shell (Radius R)	Any diameter axis.	$I = \frac{2}{3}MR^2$
Solid Cylinder (Radius R)	Central longitudinal axis.	$I = \frac{1}{2}MR^2$

Damped Harmonic Oscillator Response Regimes



Quick Revision — Module 1

Mechanics | BS-PH101

Key Formulas and Metrics at a Glance

Degrees of Freedom	$f = 3N - k$ (N particles, k constraints)
Limiting Static Friction	$f_{s,\max} = \mu_s N$ and $\tan \lambda = \mu_s$
Potential Energy Relation	$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$
Conservative Force Rule	$\nabla \times \vec{F} = 0 \iff \oint \vec{F} \cdot d\vec{r} = 0$
Coriolis Acceleration	$\vec{a}_{\text{cor}} = 2(\vec{\omega} \times \vec{v}')$ (Deflection to right in Northern Hemisphere)
Underdamped Frequency	$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ ($\gamma = b/2m$)
Quality Factor	$Q = \frac{\omega_0}{2\gamma} = \frac{\pi}{\delta} = \omega_0 \tau$
Resonance Amplitude Freq.	$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$
Parallel Axis Theorem	$I = I_{\text{cm}} + Mh^2$
Perpendicular Axis Theorem	$I_z = I_x + I_y$ (Planar lamina in xy -plane)

One-Line Definitions

- **Holonomic Constraint:** A constraint expressible as an integrable algebraic equation of position coordinates and time $f(\vec{r}, t) = 0$.
- **Conservative Force:** A force field in which work done along any closed path is zero ($\oint \vec{F} \cdot d\vec{r} = 0$).
- **Coriolis Force:** A velocity-dependent fictitious force acting in a rotating reference frame ($\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$).
- **Logarithmic Decrement:** The natural log of ratio of two consecutive amplitudes one period apart ($\delta = \gamma T_d$).
- **Radius of Gyration:** The perpendicular distance from axis of rotation where whole mass can be assumed concentrated ($I = Mk^2$).

Frequently Asked / Viva Questions

Q1. State the condition for a force field to be conservative.

A force field $\vec{F}$ is conservative if $\nabla \times \vec{F} = 0$ (irrotational) or equivalently work done around any closed loop $\oint \vec{F} \cdot d\vec{r} = 0$.

Q2. Why is Coriolis force called a fictitious force?

Frequently Asked / Viva Questions (contd.)

Because it arises due to the acceleration of the observer's rotating frame of reference, not due to physical interactions between bodies.

Q3. What is the physical significance of negative gradient of potential energy?

$\vec{F} = -\nabla V$ indicates force always points in direction of steepest decrease of potential energy towards stable equilibrium.

Q4. Distinguish between holonomic and non-holonomic constraints with examples.

Holonomic constraints are integrable algebraic position relations (e.g. rigid body length); non-holonomic involve non-integrable velocity relations or inequalities (e.g. sphere rolling without slipping).

Q5. What is critical damping and state its practical application?

Critical damping occurs when $\gamma = \omega_0$. The system returns to equilibrium in shortest time without oscillating. Used in car shock absorbers and deadbeat galvanometers.

Q6. How does amplitude resonance frequency differ from velocity resonance frequency?

Amplitude resonance occurs at $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$ while velocity resonance occurs exactly at natural frequency $\omega_v = \omega_0$.

Q7. What is Quality Factor (Q) of an oscillator?

$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma}$. Higher Q implies lower damping and sharper resonance.

Q8. State Parallel and Perpendicular axes theorems of moment of inertia.

Parallel axis: $I = I_{\text{cm}} + Mh^2$. Perpendicular axis: $I_z = I_x + I_y$ for planar laminas.

Q9. Why are track rails laid higher on outer curves?

To provide necessary centripetal force via component of normal reaction and eliminate reliance on lateral friction.

Q10. Calculate the moment of inertia of a solid sphere of mass M and radius R about a tangent.

By parallel axis theorem, $I_{\text{tangent}} = I_{\text{cm}} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$.

Q11. What is the total energy of a simple harmonic oscillator of mass m and amplitude A ?

$E = \frac{1}{2}m\omega_0^2 A^2$. Total mechanical energy is constant and continuously converts between kinetic and potential forms.

Q12. Show that Coriolis force does no work on a moving particle.

$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$ is always perpendicular to velocity $\vec{v}'$. Power $P = \vec{F}_{\text{cor}} \cdot \vec{v}' = 0 \Rightarrow W = 0$.

Summary Comparison of Damped Motion Regimes

Parameter	Overdamped ($\gamma > \omega_0$)	Critically Damped ($\gamma = \omega_0$)	Underdamped ($\gamma < \omega_0$)
Roots of Char. Eq.	Real and distinct.	Real and equal ($r = -\gamma$).	Complex conjugate ($-\gamma \pm i\omega_d$).
Motion Nature	Non-oscillatory, slow return.	Non-oscillatory, fastest return.	Oscillatory with decaying amplitude.
Frequency	No frequency of oscillation.	No frequency of oscillation.	$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$.
Practical Use	Heavy door closers.	Galvanometers, vehicle suspension.	Tuning circuits, musical strings.

Module 2

Interference vs. Diffraction and Classification

1. Interference vs. Diffraction and Classification

1.1 Distinction Between Interference and Diffraction

Definition: Interference and Diffraction

- **Interference:** Modification in intensity resulting from the superposition of two or more coherent light waves originating from distinct coherent sources.
- **Diffraction:** The bending of light waves around the sharp edges of an obstacle or aperture and their subsequent superposition within the geometrical shadow region.

Comprehensive Comparison:

Property	Interference	Diffraction
Wave Origin	Superposition of waves from two distinct coherent sources.	Superposition of secondary wavelets originating from different points of the <i>same</i> wavefront.
Fringe Width	Fringes are usually of equal width ($\beta = \lambda D/d$).	Fringes are never of equal width (central max is twice as wide as secondary max).
Intensity	All bright fringes have equal intensity ($I_{\max} = 4I_0$).	Intensity decreases rapidly with increasing order of secondary maxima.
Darkness	Minima are completely dark ($I_{\min} = 0$).	Minima are generally not completely dark.

1.2 Fraunhofer vs. Fresnel Diffraction

Feature	Fresnel Diffraction	Fraunhofer Diffraction
Distance	Source and screen are at finite distances from aperture.	Source and screen are at effectively infinite distances.
Wavefront	Incident wavefront is spherical or cylindrical.	Incident wavefront is plane (parallel rays).
Lenses	No lenses required to focus pattern.	Convex lenses required before and after aperture.
Analysis	Analyzed using Fresnel half-period zones.	Analyzed using simple wave phase summation.

2. Fraunhofer Diffraction at Single, Double, and Multiple Slits

2.1 Single Slit Diffraction

Consider a plane wavefront of wavelength λ incident normally on a slit of width e .

Single Slit Intensity Formula

The resultant intensity on a screen at diffraction angle θ is:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda} \quad (32)$$

Key Conditions:

- **Central Maximum:** At $\theta = 0 \Rightarrow \beta = 0, \lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1 \Rightarrow I = I_0$.
- **Minima Condition:** $\sin \beta = 0$ (for $\beta \neq 0$) $\Rightarrow \beta = m\pi \Rightarrow e \sin \theta = m\lambda$ ($m = \pm 1, \pm 2, \pm 3, \dots$).
- **Secondary Maxima:** Occur near $\beta = \pm 1.43\pi, \pm 2.46\pi, \dots$ with intensity ratios:

$$I_0 : I_1 : I_2 = 1 : \frac{1}{22} : \frac{1}{61} = 1 : 0.045 : 0.016 \quad (33)$$

- **Width of Central Maximum:** $\Delta\theta = \frac{2\lambda}{e} \Rightarrow$ Linear Width $W = \frac{2\lambda f}{e}$ (where f is lens focal length).

2.2 Double Slit Diffraction

For two parallel slits, each of width e and separated by opaque spacing d (grating element $= e + d$):

Double Slit Intensity Formula

$$I(\theta) = 4I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda}, \quad \alpha = \frac{\pi(e+d) \sin \theta}{\lambda} \quad (34)$$

It is the product of single-slit diffraction factor $\left(\frac{\sin \beta}{\beta} \right)^2$ and interference factor $\cos^2 \alpha$.

Missing Orders (Absent Spectra): Occurs when an interference maximum coincides with a diffraction minimum:

$$(e+d) \sin \theta = n\lambda \quad \text{and} \quad e \sin \theta = m\lambda \implies \frac{e+d}{e} = \frac{n}{m} \quad (35)$$

If $e = d$, then $\frac{n}{m} = 2 \implies n = 2m$, so 2nd, 4th, 6th interference orders disappear.

2.3 Diffraction Grating and Resolving Power

A plane transmission diffraction grating consists of N parallel, equally spaced slits with grating element $(e+d)$.

Grating Equations and Resolving Power

- Principal Maxima Condition:**

$$(e+d) \sin \theta = m\lambda \quad (m = 0, 1, 2, \dots) \quad (36)$$

- Dispersive Power (D):** Rate of change of diffraction angle with wavelength:

$$D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta} \quad (37)$$

- Resolving Power (R):** Capacity to separate two spectral lines of wavelength λ and $\lambda + d\lambda$:

$$R = \frac{\lambda}{d\lambda} = mN \quad (N = \text{total illuminated rulings}) \quad (38)$$

Solved Problem 1: Grating Spectral Lines and Maximum Order

Problem: A grating has 5000 lines/cm. Light of wavelength $\lambda = 6000 \text{ \AA}$ is incident normally. Find: (a) Grating element $(e+d)$, (b) Highest observable order.

Solution:

1. Grating element $(e+d) = \frac{1 \text{ cm}}{5000} = 2 \times 10^{-4} \text{ cm} = 2 \times 10^{-6} \text{ m}$.

2. Maximum angle $\theta_{\max} = 90^\circ \implies \sin 90^\circ = 1$.

3. Substitute in grating equation $(e+d) \sin 90^\circ = m_{\max} \lambda$:

$$m_{\max} = \frac{(e+d)}{\lambda} = \frac{2 \times 10^{-6} \text{ m}}{600 \times 10^{-9} \text{ m}} = 3.33 \implies m_{\max} = 3 \quad (39)$$

Hence, maximum 3rd order spectrum can be observed.

Solved Problem 2: Single Slit Central Width

Problem: Light of wavelength $\lambda = 5890 \text{ \AA}$ is incident normally on a slit of width 0.1 mm . A convex lens of focal length $f = 1.0 \text{ m}$ is placed behind the slit. Calculate linear width of central maximum.

Solution:

Linear width of central maximum $W = \frac{2\lambda f}{e}$. $W = \frac{2 \times 5890 \times 10^{-10} \text{ m} \times 1.0 \text{ m}}{0.1 \times 10^{-3} \text{ m}} = 11.78 \times 10^{-3} \text{ m} = 11.78 \text{ mm}$.

3. Polarisation of Light

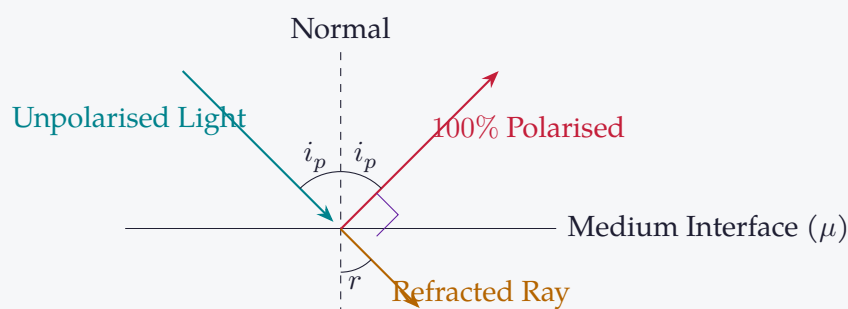
3.1 Brewster's Law and Double Refraction

Brewster's Law

When unpolarised light is incident at a specific angle i_p (polarising angle) on a dielectric interface, the reflected light is 100% plane polarised with electric vector perpendicular to plane of incidence.

$$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ \quad (40)$$

Polarisation by Reflection (Brewster's Law)



Double Refraction (Birefringence): When unpolarised light enters an anisotropic crystal (e.g. Calcite, Quartz), it splits into two plane polarised rays travelling at different speeds:

- **Ordinary Ray (O-ray):** Obeys Snell's law; travels with equal velocity in all directions ($v_o = c/\mu_o$).
- **Extraordinary Ray (E-ray):** Violates Snell's law; velocity depends on direction of propagation (v_e).

3.2 Retardation Plates: Quarter-Wave and Half-Wave Plates

Retardation Plate Thickness Equations

For a crystal plate cut parallel to optic axis of thickness t :

- **Quarter-Wave Plate (QWP):** Introduces phase difference $\Delta\phi = \pi/2$ or path

difference $\Delta x = \lambda/4$:

$$t = \frac{\lambda}{4|\mu_o - \mu_e|} \quad (41)$$

- **Half-Wave Plate (HWP):** Introduces phase difference $\Delta\phi = \pi$ or path difference $\Delta x = \lambda/2$:

$$t = \frac{\lambda}{2|\mu_o - \mu_e|} \quad (42)$$

3.3 Optical Activity and Laurent's Half-Shade Polarimeter

Optically active substances rotate the plane of polarization of linearly polarized light. The angle of rotation θ is given by Biot's law: $\theta = S \cdot L \cdot c$.

$$\text{Specific Rotation } [S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad \left(\text{in } \frac{\text{deg}}{\text{dm} \cdot \text{g/cm}^3} \right) \quad (43)$$

where θ = rotation angle (deg), L = path length in decimeters (dm), c = concentration in g/cm^3 .

Laurent's Half-Shade Polarimeter: Uses a half-shade plate (half quartz, half glass) to divide the field of view into two halves, enabling precise setting of equal brightness.

Solved Problem 3: Quarter-Wave Plate Thickness

Problem: Calculate the minimum thickness of a quarter-wave plate of Calcite for sodium light of wavelength $\lambda = 5893 \text{ \AA}$. Given $\mu_o = 1.6583$ and $\mu_e = 1.4864$.

Solution:

1. Calcite is a negative crystal ($\mu_o > \mu_e$), so $|\mu_o - \mu_e| = 1.6583 - 1.4864 = 0.1719$.
2. Formula: $t = \frac{\lambda}{4(\mu_o - \mu_e)}$.
3. Substitute: $t = \frac{5893 \times 10^{-10} \text{ m}}{4 \times 0.1719} = \frac{5893 \times 10^{-10}}{0.6876} = 8570 \times 10^{-10} \text{ m} = \mathbf{0.857 \text{ }\mu\text{m}}$.

4. Lasers (Light Amplification by Stimulated Emission of Radiation)

4.1 Principles of Laser Action: Einstein's Coefficients

Interaction of radiation with matter involves three fundamental processes:

Three Radiation Processes and Einstein's Relations

1. **Stimulated Absorption:** Atom absorbs photon: Rate $R_{12} = B_{12}N_1u(\nu)$.
2. **Spontaneous Emission:** Atom decays naturally: Rate $R_{21,\text{sp}} = A_{21}N_2$.
3. **Stimulated Emission:** Incident photon induces decay: Rate $R_{21,\text{st}} = B_{21}N_2u(\nu)$.

At thermal equilibrium, $R_{12} = R_{21,\text{sp}} + R_{21,\text{st}}$. Equating with Planck's law yields Einstein's relations:

$$B_{12} = B_{21} \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (44)$$

4.2 Population Inversion and Pumping

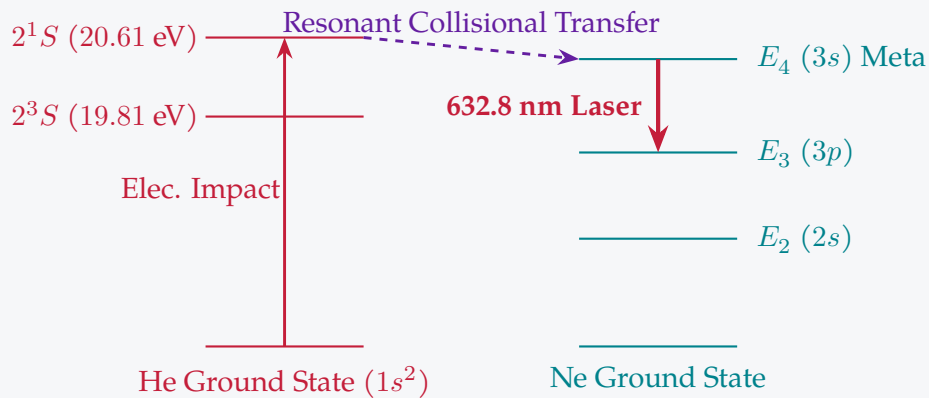
Under normal thermal equilibrium (Boltzmann distribution $N_2 = N_1 e^{-h\nu/kT}$), $N_1 \gg N_2$.

Definition: Population Inversion and Pumping

- **Population Inversion:** The non-equilibrium state in which higher energy level population exceeds lower energy level ($N_2 > N_1$). Necessary condition for optical amplification.
- **Pumping:** The external process of supplying energy to raise atoms to higher excited levels to achieve population inversion (e.g. Optical pumping, Electrical discharge, Direct electron excitation).
- **Metastable State:** An excited state with a long lifetime ($\sim 10^{-3}$ s compared to normal 10^{-8} s), allowing accumulation of atoms for population inversion.

4.3 He-Ne Laser (Gas Laser)

He-Ne Laser Energy Level Diagram



Working Mechanism of He-Ne Laser:

1. Electric discharge accelerates electrons which excite He atoms to metastable 2^1S (20.61 eV) state.
2. He atoms transfer energy to Ne atoms via resonant inelastic collisions, populating Ne's $3s$ (E_4) level.
3. Stimulated emission from $3s \rightarrow 2p$ ($E_4 \rightarrow E_3$) produces red laser light at $\lambda = 632.8$ nm.

Quick Revision — Module 2

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Key Formulas and Metrics at a Glance

Single Slit Minima	$e \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$
Single Slit Central Width	$\Delta\theta = \frac{2\lambda}{e} \Rightarrow \text{Linear Width } W = \frac{2\lambda f}{e}$
Double Slit Missing Orders	$\frac{e+d}{e} = \frac{n}{m} \quad (n = \text{interf. order}, m = \text{diff. order})$
Grating Equation	$(e + d) \sin \theta = m\lambda$
Grating Dispersive Power	$D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta}$
Grating Resolving Power	$R = \frac{\lambda}{d\lambda} = mN$
Brewster's Law	$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ$
Quarter-Wave Plate (QWP)	$t = \frac{\lambda}{4 \mu_o - \mu_e }$
Half-Wave Plate (HWP)	$t = \frac{\lambda}{2 \mu_o - \mu_e }$
Specific Rotation	$[S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad (\text{deg} \cdot \text{dm}^{-1} \cdot (\text{g/cm}^3)^{-1})$
Einstein Coefficients Relation	$\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \quad \text{and} \quad B_{12} = B_{21}$

One-Line Definitions

- **Diffraction Grating:** An optical component with a periodic structure of N slits that diffracts light into several beams.
- **Brewster Angle (i_p):** Angle of incidence at which light with a particular polarization is perfectly transmitted without reflection.
- **Birefringence:** The optical property of a material having a refractive index that depends on the polarization and propagation direction of light.
- **Population Inversion:** State in which a system exists with more members in an excited state than in lower energy states ($N_2 > N_1$).
- **Metastable State:** An excited atomic state with a long lifetime ($\sim 10^{-3}$ s), essential for building up population inversion.

Frequently Asked / Viva Questions

Q1. What is the basic difference between interference and diffraction?

Interference results from superposition of waves from two distinct coherent sources; diffraction results from superposition of wavelets from different parts of the same wavefront.

Frequently Asked / Viva Questions (contd.)

Q2. Why is central maximum in single-slit diffraction twice as wide as secondary maxima?

Because central maximum extends between first minima on either side ($\sin \theta = \pm \lambda/e$), spanning angular width $2\lambda/e$, whereas secondary maxima span width λ/e .

Q3. What is meant by missing orders in double slit diffraction?

Missing orders occur when an interference maximum coincides with a diffraction minimum at the same angle θ , making that order absent ($\frac{e+d}{e} = \frac{n}{m}$).

Q4. State Rayleigh's criterion for spectral resolution.

Two spectral lines are just resolved when the central principal maximum of one falls on the first minimum of the other.

Q5. How does resolving power of a grating differ from dispersive power?

Dispersive power ($d\theta/d\lambda$) measures angular separation between lines; resolving power ($\lambda/d\lambda = mN$) measures ability to distinguish two close wavelengths.

Q6. State Brewster's law and show reflected and refracted rays are perpendicular.

$\mu = \tan i_p$. Since $\frac{\sin i_p}{\sin r} = \tan i_p = \frac{\sin i_p}{\cos i_p} \Rightarrow \sin r = \cos i_p = \sin(90^\circ - i_p) \Rightarrow i_p + r = 90^\circ$.

Q7. What are O-ray and E-ray in a birefringent crystal?

O-ray (Ordinary) obeys Snell's law with constant speed v_o ; E-ray (Extraordinary) violates Snell's law with directional speed v_e .

Q8. Distinguish between positive and negative uniaxial crystals.

In positive crystals ($\mu_e > \mu_o$, e.g. Quartz), $v_o > v_e$; in negative crystals ($\mu_o > \mu_e$, e.g. Calcite), $v_e > v_o$.

Q9. Explain the function of a Quarter-Wave Plate (QWP).

A QWP introduces a path difference of $\lambda/4$ ($\pi/2$ phase shift) between O-ray and E-ray, converting linearly polarized light to circularly or elliptically polarized light.

Q10. What is population inversion and why is it necessary for laser action?

Population inversion ($N_2 > N_1$) ensures that rate of stimulated emission exceeds rate of stimulated absorption, producing net optical amplification.

Q11. Why is Helium gas added to Neon gas in a He-Ne laser?

He atoms are excited by electron impact to metastable states ($2^1S, 2^3S$) and efficiently transfer energy to Ne atoms via resonant collisions, achieving population inversion in Ne.

Q12. Derive relation between Einstein A and B coefficients.

Equating thermal equilibrium rate $B_{12}N_1u(\nu) = A_{21}N_2 + B_{21}N_2u(\nu)$ with Boltzmann distribution $N_2/N_1 = e^{-h\nu/kT}$ and Planck's radiation law gives $B_{12} = B_{21}$ and $A_{21}/B_{21} = 8\pi h\nu^3/c^3$.

Summary Comparison of Laser Processes

Process	Incident Condition	Transition	Output Properties
Absorption	Photon ($h\nu = E_2 - E_1$)	Lower $E_1 \rightarrow$ Higher E_2	Atom reaches excited state.
Spontaneous Emission	Natural decay (lifetime $\sim 10^{-8}\text{s}$)	Higher $E_2 \rightarrow$ Lower E_1	Incoherent, random phase/direction.
Stimulated Emission	Triggering photon ($h\nu$)	Higher $E_2 \rightarrow$ Lower E_1	Coherent, identical phase, direction, and wavelength.

Module 3

Maxwell's Equations and Electromagnetic Waves

1. Maxwell's Equations and Electromagnetic Waves

1.1 Maxwell's Four Fundamental Equations

Maxwell unified electricity and magnetism into four differential and integral equations governing electromagnetic phenomena.

Maxwell's Equations in Differential and Integral Forms

1. **Gauss's Law for Electrostatics:** Net electric flux through a closed surface equals charge enclosed:

$$\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}} \quad (\vec{D} = \epsilon_0 \vec{E} + \vec{P}) \quad (45)$$

2. **Gauss's Law for Magnetostatics:** Magnetic monopoles do not exist (magnetic flux lines are continuous loops):

$$\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0 \quad (46)$$

3. **Faraday's Law of Electromagnetic Induction:** Time-varying magnetic field produces an electric field:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{A} \quad (47)$$

4. **Ampere-Maxwell Law:** Magnetic fields are generated by electric currents and time-varying electric fields:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \iff \oint_C \vec{H} \cdot d\vec{r} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{A} \quad (48)$$

1.2 Displacement Current and Wave Equation in Free Space

Maxwell modified Ampere's law ($\nabla \times \vec{H} = \vec{J}$) by taking the divergence of both sides. Since $\nabla \cdot (\nabla \times \vec{H}) = 0$, it required $\nabla \cdot \vec{J} = 0$, which violates the charge continuity equation

$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$ for time-varying fields.

Substituting $\rho = \nabla \cdot \vec{D}$ into continuity equation:

$$\nabla \cdot \vec{J} + \frac{\partial}{\partial t}(\nabla \cdot \vec{D}) = 0 \implies \nabla \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0 \quad (49)$$

Thus, Maxwell introduced the **Displacement Current Density** $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$.

Electromagnetic Wave Equation and Speed of Light

In free space ($\rho = 0, \vec{J} = 0, \mu = \mu_0, \epsilon = \epsilon_0$), taking curl of Faraday's law yields:

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}) \implies \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad (50)$$

Since $\nabla \cdot \vec{E} = 0$, this simplifies to the standard 3D wave equation:

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{and} \quad \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad (51)$$

Comparing with $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$, the wave velocity in vacuum is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}} = 2.998 \times 10^8 \text{ m/s} \quad (52)$$

1.3 Poynting Vector and Energy Flow

Definition: Poynting Vector

The **Poynting Vector** ($\vec{S}$) represents the directional energy flux density (rate of energy transfer per unit area) of an electromagnetic field:

$$\vec{S} = \vec{E} \times \vec{H} \quad \left(\text{in } \frac{\text{W}}{\text{m}^2} \right) \quad (53)$$

The time-averaged Poynting vector intensity for a plane wave is $I = \langle S \rangle = \frac{1}{2} c \epsilon_0 E_0^2$.

Solved Problem 1: Displacement Current in a Capacitor

Problem: A parallel-plate capacitor with circular plates of radius $R = 0.1 \text{ m}$ is being charged by a current $I = 2.0 \text{ A}$. Calculate: (a) Displacement current I_d , (b) Rate of change of electric field dE/dt .

Solution:

1. Total displacement current between plates equals conduction current in wires:
 $I_d = 2.0 \text{ A}$.
2. Plate area $A = \pi R^2 = \pi(0.1)^2 = 0.0314 \text{ m}^2$.

$$3. I_d = \epsilon_0 A \frac{dE}{dt} \Rightarrow \frac{dE}{dt} = \frac{I_d}{\epsilon_0 A}:$$

$$\frac{dE}{dt} = \frac{2.0}{(8.854 \times 10^{-12}) \times 0.0314} = 7.19 \times 10^{11} \text{ V/(m} \cdot \text{s)} \quad (54)$$

2. Dielectric Properties of Materials

2.1 Dielectric Vectors: Polarisation, Susceptibility, Permittivity

Dielectrics are electrical insulators that can be polarized by an applied electric field.

Fundamental Dielectric Relations

- **Electric Dipole Moment ($\vec{p}$):** $\vec{p} = q\vec{d}$ (directed from $-q$ to $+q$).
- **Polarisation Vector ($\vec{P}$):** Dipole moment per unit volume: $\vec{P} = N\vec{p}$.
- **Electric Displacement ($\vec{D}$):**

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \epsilon_r \vec{E} \quad (55)$$

- **Electric Susceptibility (χ_e):** $\vec{P} = \epsilon_0 \chi_e \vec{E} \Rightarrow \epsilon_r = 1 + \chi_e$.

Microscopic Polarisation Mechanisms:

Mechanism	Origin / Description	Temperature Dependence
Electronic (α_e)	Displacement of electron cloud relative to nucleus in an atom.	Independent of temperature (T).
Ionic (α_i)	Relative displacement of positive and negative ions in a crystal lattice.	Independent of temperature (T).
Orientational (α_o)	Realignment of permanent molecular dipoles along external field.	Strongly temperature dependent ($\alpha_o \propto 1/T$).
Space-Charge (α_s)	Accumulation of free charge carriers at interfaces/grain boundaries.	Dependent on temperature and frequency.

2.2 Clausius-Mossotti Equation

Consider a solid dielectric where local field acting on an atom is the **Lorentz Field**:

$$E_{\text{local}} = E + \frac{P}{3\epsilon_0} \quad (56)$$

The induced dipole moment per atom is $p = \alpha E_{\text{local}}$, so total polarisation for N atoms per unit volume is:

$$P = Np = N\alpha E_{\text{local}} = N\alpha \left(E + \frac{P}{3\epsilon_0} \right) \quad (57)$$

Clausius-Mossotti Relation Derivation

Solving for P :

$$P \left(1 - \frac{N\alpha}{3\epsilon_0} \right) = N\alpha E \Rightarrow P = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \quad (58)$$

Since $P = (\epsilon_r - 1)\epsilon_0 E$, substitute for P :

$$(\epsilon_r - 1)\epsilon_0 E = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \Rightarrow \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} \quad (59)$$

This connects the macroscopic dielectric constant ϵ_r with the microscopic atomic polarizability α .

3. Magnetic Properties of Materials

3.1 Magnetic Vectors and Classification

Magnetic Field Relations

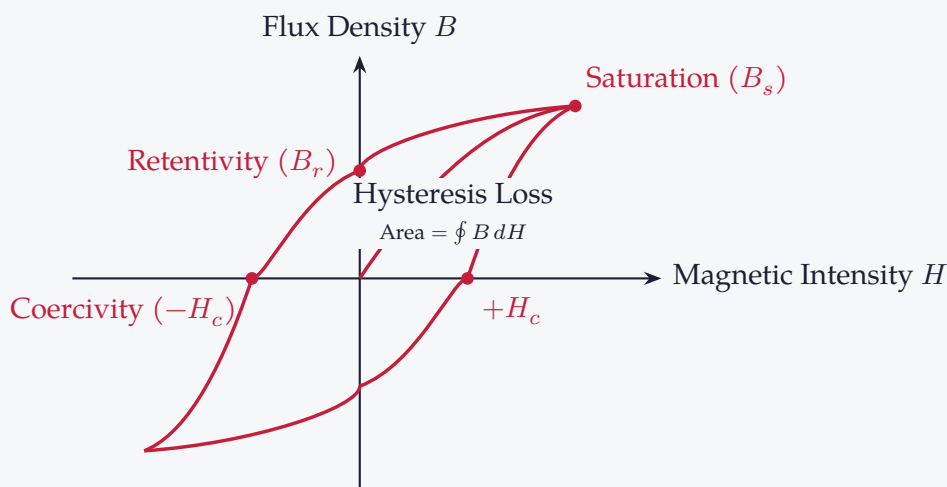
- **Magnetisation ($\vec{M}$):** Magnetic dipole moment per unit volume: $\vec{M} = \frac{\sum \vec{m}}{V}$.
- **Magnetic Flux Density ($\vec{B}$):** $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0\mu_r\vec{H}$.
- **Magnetic Susceptibility (χ_m):** $\vec{M} = \chi_m\vec{H} \Rightarrow \mu_r = 1 + \chi_m$.

Classification of Magnetic Materials:

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Atomic Dipoles	No permanent dipoles.	Permanent dipoles present (random).	Permanent dipoles strongly aligned in domains.
Susceptibility χ_m	Small and negative ($\sim -10^{-5}$).	Small and positive ($\sim 10^{-4}$).	Very large and positive ($\sim 10^3 - 10^5$).
Permittivity μ_r	$\mu_r < 1$ slightly.	$\mu_r > 1$ slightly.	$\mu_r \gg 1$.
Temp. Dependence	Independent of T .	Curie Law ($\chi \propto 1/T$).	Curie-Weiss Law ($\chi \propto \frac{1}{T-T_c}$).
External Field	Repelled by magnetic fields.	Feebly attracted to magnetic fields.	Strongly attracted into magnetic fields.
Examples	Bismuth, Copper, Water, Gold.	Aluminium, Platinum, Oxygen.	Iron, Cobalt, Nickel, Gadolinium.

3.2 Hysteresis Loop (B-H Curve)

Ferromagnetic Hysteresis (B-H) Loop



Hysteresis Terminology

- **Retentivity (Remanence B_r):** Residual flux density remaining in material when magnetising field H is reduced to zero.
- **Coercivity (H_c):** Reverse magnetising field required to reduce residual flux density B to zero.
- **Hysteresis Loss:** Energy dissipated per unit volume per cycle as heat during magnetisation reversal, equal to area enclosed by B - H loop: $W_{\text{loss}} = \oint B dH$.

Soft vs. Hard Magnetic Materials:

Feature	Soft Magnetic Materials	Hard Magnetic Materials
Hysteresis Area	Thin and narrow loop (low energy loss).	Broad and wide loop (high energy loss).
Coercivity	Very low coercivity (H_c small).	Very high coercivity (H_c large).
Retentivity	High retentivity, easily demagnetised.	High retentivity, retains magnetisation.
Applications	Transformer cores, electromagnets, inductors.	Permanent magnets, magnetic recording tapes, loudspeakers.

Solved Problem 2: Hysteresis Loss Power Dissipation

Problem: An iron ring of volume $V = 100 \text{ cm}^3$ is subjected to a 50 Hz alternating magnetic field. If the area of the hysteresis loop corresponds to 250 J/m^3 per cycle, calculate the power loss in watts.

Solution:

1. Volume $V = 100 \times 10^{-6} \text{ m}^3 = 10^{-4} \text{ m}^3$.
2. Frequency $f = 50 \text{ Hz}$ (50 cycles/second).

3. Energy loss per second (Power P) = Energy/cycle/volume $\times$ Volume $\times$ Frequency:

$$P = W_{\text{loss}} \times V \times f = 250 \text{ J/m}^3 \times 10^{-4} \text{ m}^3 \times 50 \text{ s}^{-1} = \mathbf{1.25 \text{ W}} \quad (60)$$

Quick Revision — Module 3

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Key Formulas and Metrics at a Glance

Maxwell 1 (Gauss Electro)	$\nabla \cdot \vec{D} = \rho \iff \oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}$
Maxwell 2 (Gauss Magneto)	$\nabla \cdot \vec{B} = 0 \iff \oint_S \vec{B} \cdot d\vec{A} = 0$
Maxwell 3 (Faraday)	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \iff \oint_C \vec{E} \cdot d\vec{r} = -\frac{\partial \Phi_B}{\partial t}$
Maxwell 4 (Ampere-Maxwell)	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
Displacement Current	$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$
Speed of Light in Vacuum	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$
Poynting Vector	$\vec{S} = \vec{E} \times \vec{H} \quad (\text{W/m}^2)$
Clausius-Mossotti Eq.	$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$
Magnetic Relations	$\vec{B} = \mu_0(\vec{H} + \vec{M}) \quad \text{and} \quad \mu_r = 1 + \chi_m$
Hysteresis Power Loss	$P_{\text{loss}} = (\oint B dH) \times \text{Volume} \times \text{Frequency}$

One-Line Definitions

- **Displacement Current:** A quantity appearing in Maxwell's equations defined by rate of change of electric displacement field ($\vec{J}_d = \partial \vec{D} / \partial t$).
- **Poynting Vector:** A vector representing the directional energy flux density of an electromagnetic field ($\vec{S} = \vec{E} \times \vec{H}$).
- **Lorentz Local Field:** The actual local electric field acting at the position of an individual atom in a solid ($E_{\text{local}} = E + P/3\epsilon_0$).
- **Retentivity (B_r):** Residual magnetic flux density remaining in a ferromagnetic material when magnetising field H is reduced to zero.
- **Coercivity (H_c):** The magnitude of reverse magnetic field intensity required to reduce residual magnetisation to zero.

Frequently Asked / Viva Questions

Q1. Why did Maxwell modify Ampere's law and what term did he add?

Ampere's law ($\nabla \times \vec{H} = \vec{J}$) violated charge conservation for time-varying fields ($\nabla \cdot \vec{J} \neq 0$). Maxwell added displacement current density $\vec{J}_d = \partial \vec{D} / \partial t$.

Q2. What is the physical significance of $\nabla \cdot \vec{B} = 0$?

Frequently Asked / Viva Questions (contd.)

It signifies that isolated magnetic monopoles do not exist in nature and magnetic flux lines always form continuous closed loops.

Q3. What is Poynting vector and state its SI unit?

$\vec{S} = \vec{E} \times \vec{H}$ represents the rate of electromagnetic energy flow per unit area. Its SI unit is W/m^2 .

Q4. State the relation between dielectric constant ϵ_r and electric susceptibility χ_e .

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0(1 + \chi_e)\vec{E} = \epsilon_0 \epsilon_r \vec{E} \implies \epsilon_r = 1 + \chi_e.$$

Q5. Derive Clausius-Mossotti relation connecting macroscopic and microscopic dielectric properties.

Using Lorentz local field $E_{\text{local}} = E + P/3\epsilon_0$, total polarisation $P = N\alpha E_{\text{local}} = N\alpha(E + P/3\epsilon_0)$. Substituting $P = (\epsilon_r - 1)\epsilon_0 E$ yields $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$.

Q6. Distinguish between polar and non-polar dielectrics.

Polar dielectrics have permanent molecular dipole moments (e.g. H_2O , HCl); non-polar dielectrics have zero permanent dipole moment in zero field (e.g. O_2 , N_2 , CO_2).

Q7. Why does orientational polarizability decrease with increasing temperature?

Because higher thermal agitation randomizes the orientation of permanent molecular dipoles, opposing their alignment along the external field ($\alpha_o \propto 1/T$).

Q8. Distinguish between diamagnetic and paramagnetic materials.

Diamagnetic materials have no permanent dipoles, negative susceptibility ($\chi_m < 0$), and are repelled by magnetic fields. Paramagnetic materials have permanent dipoles, positive susceptibility ($\chi_m > 0$), and are feebly attracted.

Q9. State Curie-Weiss law for ferromagnetic materials.

Above the Curie temperature T_c , a ferromagnetic material becomes paramagnetic with susceptibility $\chi_m = \frac{C}{T - T_c}$.

Q10. What is meant by hysteresis loop area?

The area enclosed by the B - H loop represents the energy dissipated as heat per unit volume per magnetization reversal cycle ($W_{\text{loss}} = \oint B dH$).

Q11. Which magnetic material is suitable for transformer cores and why?

Soft magnetic materials (e.g. Soft Iron, Silicon Steel) because they have low coercivity and thin hysteresis loops, minimizing energy loss per cycle.

Q12. Show that speed of EM wave in vacuum is $c = 1/\sqrt{\mu_0 \epsilon_0}$.

From Maxwell's free-space wave equation $\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$, comparing with standard wave equation $\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ yields $v = c = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \times 10^8 \text{ m/s}$.

Summary Comparison of Dielectric Polarization Mechanisms

Mechanism	Displacement Type	Frequency Range	Temperature Dependence
Electronic (α_e)	Electron cloud relative to nucleus.	Optical ($\sim 10^{15}$ Hz).	Independent of T .
Ionic (α_i)	Positive and negative ion displacement.	Infrared ($\sim 10^{13}$ Hz).	Independent of T .
Orientational (α_o)	Alignment of permanent dipoles.	Microwave ($\sim 10^9$ Hz).	Inverse with T ($\alpha_o \propto 1/T$).
Space-Charge (α_s)	Migration of charge carriers.	Audio ($\sim 10^3$ Hz).	Complex T dependence.

Module 4

Inadequacy of Classical Physics and Black Body Radiation

1. Inadequacy of Classical Physics and Black Body Radiation

1.1 Failure of Classical Mechanics and Planck's Radiation Law

Classical physics (Rayleigh-Jeans law $u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$) predicted that black body radiation energy density tends to infinity at short wavelengths ($\lambda \rightarrow 0$), known as the **Ultraviolet Catastrophe**.

Planck's Quantum Hypothesis (1900)

Black body cavity walls consist of atomic oscillators that absorb or emit electromagnetic radiation in discrete packets called **quanta** (photons) of energy:

$$E = nh\nu \quad (n = 1, 2, 3, \dots) \quad (61)$$

where $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ is Planck's constant.

Planck's Black Body Radiation Law Derivation

Average energy of an oscillator $\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu e^{-nh\nu/k_B T}}{\sum_{n=0}^{\infty} e^{-nh\nu/k_B T}} = \frac{h\nu}{e^{h\nu/k_B T} - 1}$. Multiplying by number of modes per unit volume $N(\nu)d\nu = \frac{8\pi\nu^2}{c^3}d\nu$:

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu \iff u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda \quad (62)$$

Deduction of Limiting Laws:

- **Rayleigh-Jeans Law (Long λ):** For $h\nu \ll k_B T$, $e^{h\nu/k_B T} - 1 \approx \frac{h\nu}{k_B T} \implies u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$.
- **Wien's Distribution Law (Short λ):** For $h\nu \gg k_B T$, $e^{h\nu/k_B T} - 1 \approx e^{h\nu/k_B T} \implies u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda k_B T} d\lambda$.

2. Compton Effect, de Broglie Hypothesis, and Matter Waves

2.1 Compton Effect

When a high-energy X-ray or γ -ray photon collides elastically with a stationary free electron, the scattered photon emerges with a longer wavelength ($\lambda' > \lambda$).

Derivation of Compton Shift Formula

Applying conservation of relativistic energy and momentum during collision:

$$\text{Energy Conservation: } h\nu + m_0c^2 = h\nu' + mc^2 \quad (63)$$

$$\text{Momentum Conservation: } \vec{p}_{\text{photon}} = \vec{p}'_{\text{photon}} + \vec{p}_{\text{electron}} \quad (64)$$

Eliminating electron velocity v using $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$ yields the **Compton Shift Equation**:

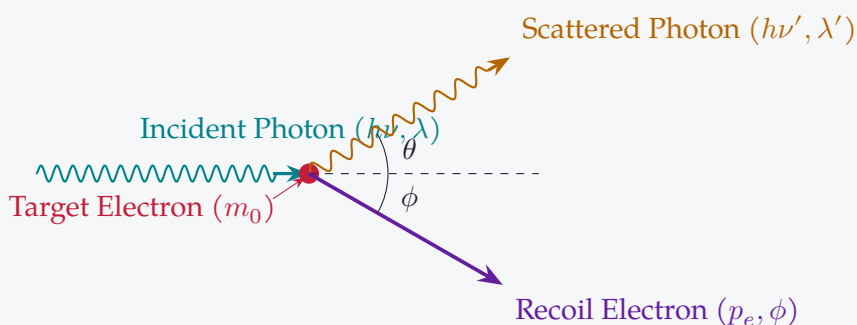
$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c}(1 - \cos\theta) = 2\lambda_c \sin^2\left(\frac{\theta}{2}\right) \quad (65)$$

where $\lambda_c = \frac{h}{m_0c} = 0.02426 \text{ \AA} = 2.426 \times 10^{-12} \text{ m}$ is the **Compton Wavelength**.

Key Characteristics of Compton Shift:

- $\Delta\lambda$ depends *only* on scattering angle θ , independent of incident wavelength or target material.
- Maximum shift occurs at $\theta = 180^\circ$ (backscattering): $\Delta\lambda_{\text{max}} = \frac{2h}{m_0c} = 0.0485 \text{ \AA}$.
- Compton shift is absent for visible light because $\Delta\lambda \ll \lambda_{\text{visible}}$ is unresolvable.

Compton Scattering Geometry



2.2 de Broglie Hypothesis and Davisson-Germer Experiment

Definition: de Broglie Hypothesis

Louis de Broglie (1924) proposed that moving matter particles exhibit wave properties with a wavelength (λ) inversely proportional to momentum (p):

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}} \quad (66)$$

For an electron accelerated through potential V volts:

$$\lambda_e = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (67)$$

Davisson-Germer Experiment: Accelerated electrons at $V = 54$ V onto a Nickel crystal target observed a pronounced scattering peak at angle $\phi = 50^\circ$. Bragg diffraction angle $\theta = 90^\circ - \phi/2 = 65^\circ$.

$$\lambda_{\text{Bragg}} = 2d \sin \theta = 2(0.91 \text{ \AA}) \sin 65^\circ = 1.65 \text{ \AA} \quad \text{vs} \quad \lambda_{\text{de Broglie}} = \frac{12.27}{\sqrt{54}} = 1.67 \text{ \AA} \quad (68)$$

This excellent agreement provided direct experimental verification of matter waves.

Solved Problem 1: Compton Shift at 90 Degrees

Problem: X-rays of wavelength $\lambda = 1.0 \text{ \AA}$ are scattered by a target at $\theta = 90^\circ$. Calculate: (a) Scattered wavelength λ' , (b) Energy of recoil electron.

Solution:

1. $\Delta\lambda = \lambda_c(1 - \cos 90^\circ) = 0.02426 \text{ \AA}(1 - 0) = 0.02426 \text{ \AA}$.
2. Scattered wavelength $\lambda' = \lambda + \Delta\lambda = 1.0 + 0.02426 = 1.02426 \text{ \AA}$.
3. Recoil electron kinetic energy $K_e = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = hc \frac{\Delta\lambda}{\lambda\lambda'}$:

$$K_e = \frac{(6.626 \times 10^{-34})(3 \times 10^8)(0.02426 \times 10^{-10})}{(1.0 \times 10^{-10})(1.02426 \times 10^{-10})} = 4.706 \times 10^{-17} \text{ J} = 294.1 \text{ eV} \quad (69)$$

3. Heisenberg Uncertainty Principle

3.1 Uncertainty Relations

Heisenberg Uncertainty Principle Relations

It is impossible to simultaneously measure both the exact position and exact momentum of a particle with absolute precision:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \quad \left(\text{where } \hbar = \frac{h}{2\pi} = 1.054 \times 10^{-34} \text{ J} \cdot \text{s} \right) \quad (70)$$

Alternate canonical conjugate forms:

$$\text{Energy-Time Uncertainty: } \Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (71)$$

$$\text{Angular Position-Momentum: } \Delta \theta \cdot \Delta L_z \geq \frac{\hbar}{2} \quad (72)$$

3.2 Applications: Non-Existence of Electron in Nucleus

Proof: Non-Existence of Electron in Nucleus

Premise: Radius of atomic nucleus is $r \approx 5 \times 10^{-15}$ m. If an electron exists inside, maximum position uncertainty $\Delta x \approx 10^{-14}$ m.

1. Minimum momentum uncertainty:

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.054 \times 10^{-34}}{2 \times 10^{-14}} = 5.27 \times 10^{-21} \text{ kg} \cdot \text{m/s} \quad (73)$$

2. Minimum relativistic energy of electron ($E \approx pc$ for $p \gg m_0c$):

$$E_{\min} \approx \Delta p \cdot c = (5.27 \times 10^{-21}) \times (3 \times 10^8) = 1.58 \times 10^{-12} \text{ J} \approx 9.9 \text{ MeV} \quad (74)$$

3. Experimental β -decay electrons emitted from nucleus have maximum energies of only 2 – 3 MeV.

Conclusion: Electrons **cannot reside** inside the nucleus.

4. Schrödinger Wave Equation and Wave Function

4.1 Physical Interpretation of Wave Function ψ

The wave function $\psi(\vec{r}, t)$ describes the quantum state of a particle.

Born's Probability Interpretation of Wave Function

ψ itself is a complex quantity without direct physical measurement. However, the scalar product $|\psi(\vec{r}, t)|^2 = \psi^* \psi$ represents the **probability density** of finding the particle at position $\vec{r}$ at time t .

$$P = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1 \quad (\text{Normalization Condition}) \quad (75)$$

Criteria for an Acceptable Quantum Wave Function:

1. ψ must be **single-valued** at all points in space.
2. ψ and its spatial derivative $\frac{\partial \psi}{\partial x}$ must be **continuous** everywhere.

3. ψ must be **square-integrable** (normalizable, $\psi \rightarrow 0$ as $r \rightarrow \infty$).

4.2 Time-Dependent and Time-Independent Schrödinger Equations

Schrödinger Wave Equations

- **1D Time-Dependent Schrödinger Equation:**

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x, t) \psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t} \quad (76)$$

- **1D Time-Independent Schrödinger Equation** (for stationary states $\psi(x, t) = \psi(x)e^{-iEt/\hbar}$):

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V(x)) \psi(x) = 0 \iff \hat{H}\psi = E\psi \quad (77)$$

where $\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V$ is the Hamiltonian operator.

5. Quantum Applications: Particle in Box, Harmonic Oscillator, H-Atom

5.1 Particle in a 1D Infinite Potential Box

Consider a particle of mass m confined in a 1D box of length L ($V(x) = 0$ for $0 < x < L$, and $V(x) = \infty$ outside).

Inside the box ($0 < x < L$), $\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$ where $k = \frac{\sqrt{2mE}}{\hbar}$. Boundary conditions $\psi(0) = 0$ and $\psi(L) = 0$ require $kL = n\pi$.

1D Box Eigenvalues and Normalised Eigenfunctions

$$\text{Energy Eigenvalues: } E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2} \quad (n = 1, 2, 3, \dots) \quad (78)$$

$$\text{Normalised Wavefunctions: } \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (79)$$

Key Consequences:

- **Energy Quantization:** Energy levels are discrete and proportional to n^2 .
- **Zero-Point Energy (E_1):** The lowest possible energy is $E_1 = \frac{h^2}{8mL^2} \neq 0$, agreeing with uncertainty principle.
- **Degeneracy in 3D Cubic Box:** $E_{n_x, n_y, n_z} = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)$. States with different quantum numbers sharing same energy are **degenerate** (e.g. $(1, 1, 2), (1, 2, 1), (2, 1, 1)$ is 3-fold degenerate).

5.2 Quantum Harmonic Oscillator and Hydrogen Atom

Quantum Harmonic Oscillator and Hydrogen Atom Energies

- **Quantum Harmonic Oscillator** ($V = \frac{1}{2}m\omega^2x^2$):

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (n = 0, 1, 2, \dots) \quad (80)$$

Zero-point energy $E_0 = \frac{1}{2}\hbar\omega$ (unlike classical oscillator which can rest at zero energy).

- **Hydrogen Atom (Coulomb Potential $V = -\frac{e^2}{4\pi\epsilon_0 r}$):**

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2n^2} = -\frac{13.6}{n^2} \text{ eV} \quad (n = 1, 2, 3, \dots) \quad (81)$$

Described by four quantum numbers: Principal (n), Orbital (l), Magnetic (m_l), and Spin (m_s).

Solved Problem 2: Particle in 1D Box Transition Energy

Problem: An electron is trapped in a 1D infinite box of width $L = 1.0 \text{ \AA}$. Calculate: (a) Ground state energy E_1 in eV, (b) Wavelength of photon emitted during transition from $n = 2 \rightarrow n = 1$.

Solution:

1. Ground state energy $E_1 = \frac{h^2}{8mL^2}$:

$$E_1 = \frac{(6.626 \times 10^{-34})^2}{8 \times (9.11 \times 10^{-31}) \times (1.0 \times 10^{-10})^2} = 6.025 \times 10^{-18} \text{ J} = \mathbf{37.66 \text{ eV}} \quad (82)$$

2. First excited energy $E_2 = 2^2 E_1 = 4(37.66) = 150.64 \text{ eV}$.

3. Energy difference $\Delta E = E_2 - E_1 = 3E_1 = 112.98 \text{ eV} = 1.808 \times 10^{-17} \text{ J}$.

4. Emitted wavelength $\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{1.808 \times 10^{-17}} = 1.10 \times 10^{-8} \text{ m} = \mathbf{110 \text{ \AA}}$.

Quick Revision — Module 4

Quantum Mechanics | BS-PH101

Key Formulas and Metrics at a Glance

Planck Photon Energy	$E = h\nu = \frac{hc}{\lambda}$
Planck Radiation Law	$u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda$
Compton Shift Equation	$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\theta)$
Compton Wavelength	$\lambda_c = \frac{h}{m_0 c} = 0.02426 \text{ \AA}$
de Broglie Wavelength	$\lambda = \frac{h}{p} = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (\text{for electron})$
Heisenberg Uncertainty	$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad \text{and} \quad \Delta E \cdot \Delta t \geq \frac{\hbar}{2}$
1D Time-Indep. Schrödinger	$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$
1D Box Energy Eigenvalue	$E_n = \frac{n^2 h^2}{8mL^2} \quad (n = 1, 2, 3, \dots)$
1D Box Wavefunction	$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$
Harmonic Oscillator Energy	$E_n = (n + 1/2)\hbar\omega \quad (\text{Zero-point } E_0 = \frac{1}{2}\hbar\omega)$
Hydrogen Atom Energy	$E_n = -\frac{13.6}{n^2} \text{ eV}$

One-Line Definitions

- **Compton Effect:** The increase in wavelength of X-ray/ γ -ray photons when scattered by weakly bound target electrons.
- **Wave-Particle Duality:** The property of matter and light exhibiting both wave-like and particle-like behaviors depending on experimental setup.
- **Probability Density ($|\psi|^2$):** The probability per unit volume of finding a quantum particle at a specific spatial coordinate.
- **Zero-Point Energy:** The lowest possible non-zero ground-state energy that a quantum mechanical physical system may possess ($E_1 \neq 0$).
- **Degeneracy:** The phenomenon where two or more distinct quantum states share the exact same energy eigenvalue.

Frequently Asked / Viva Questions

Q1. What was Rayleigh-Jeans law and why did it fail?

Rayleigh-Jeans law ($u(\nu) \propto \nu^2 T$) predicted infinite energy density at high frequencies ($\lambda \rightarrow 0$), causing the ultraviolet catastrophe.

Frequently Asked / Viva Questions (contd.)

Q2. State Planck's quantum hypothesis.

Atomic oscillators absorb or emit electromagnetic radiation only in discrete energy quanta $E = nh\nu$.

Q3. Why does Compton effect observe no shift at $\theta = 0^\circ$ and maximum shift at $\theta = 180^\circ$?

$\Delta\lambda = \lambda_c(1 - \cos\theta)$. At $\theta = 0^\circ$, $\cos 0^\circ = 1 \Rightarrow \Delta\lambda = 0$. At $\theta = 180^\circ$, $\cos 180^\circ = -1 \Rightarrow \Delta\lambda = 2\lambda_c = 0.0485 \text{ \AA}$.

Q4. Why is Compton effect not observable with visible light?

Visible light has wavelength $\sim 5000 \text{ \AA}$. The maximum Compton shift ($0.0485 \text{ \AA}$) is less than 0.001% of λ_{visible} and cannot be resolved.

Q5. What is de Broglie wavelength of an electron accelerated through V volts?

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{12.27}{\sqrt{V}} \text{ \AA}.$$

Q6. Explain the physical significance of Davisson-Germer experiment.

It demonstrated electron diffraction from a Nickel crystal lattice, proving experimentally that moving material particles possess wave characteristics.

Q7. Prove using uncertainty principle that electron cannot reside inside atomic nucleus.

For nuclear radius $\Delta x \approx 10^{-14} \text{ m}$, $\Delta p \geq \hbar/2\Delta x \Rightarrow E \approx \Delta p \cdot c \approx 9.9 \text{ MeV}$. Experimental β -decay energies are only $2 - 3 \text{ MeV}$, proving electrons cannot reside in nucleus.

Q8. State Born's physical interpretation of wave function ψ .

ψ itself has no direct physical measurement, but $|\psi(\vec{r}, t)|^2 = \psi^*\psi$ represents the probability density of finding the particle at $\vec{r}$ at time t .

Q9. What are the mandatory conditions for an acceptable wave function?

ψ must be single-valued, continuous everywhere along with its first spatial derivative, and square-integrable ($\int |\psi|^2 dV = 1$).

Q10. Why is ground state energy of particle in 1D box non-zero?

If $E_1 = 0 \Rightarrow p = 0 \Rightarrow \Delta p = 0$, requiring $\Delta x = \infty$ by uncertainty principle, contradicting confinement in box of finite length L .

Q11. What is zero-point energy of a quantum harmonic oscillator?

$E_0 = \frac{1}{2}\hbar\omega$. Unlike classical oscillators, quantum oscillators retain non-zero vibrational energy even at absolute zero temperature ($T = 0 \text{ K}$).

Q12. What is meant by degeneracy of energy levels?

Degeneracy occurs when multiple independent quantum wavefunctions (different sets of quantum numbers) correspond to the exact same energy eigenvalue (e.g. $(1, 1, 2)$, $(1, 2, 1)$, $(2, 1, 1)$ in a 3D box).

Summary Comparison of Quantum Models

System Model	Potential $V(x)$	Ground State Energy (E_1 or E_0)	Energy Spacing (ΔE)
Particle in 1D Box	$V = 0$ ($0 < x < L$), $V = \infty$ outside	$E_1 = \frac{h^2}{8mL^2}$	Non-uniform ($\propto n^2$).
Harmonic Oscillator	$V = \frac{1}{2}m\omega^2 x^2$	$E_0 = \frac{1}{2}\hbar\omega$	Uniform ($\Delta E = \hbar\omega$).
Hydrogen Atom	$V = -\frac{e^2}{4\pi\epsilon_0 r}$	$E_1 = -13.6 \text{ eV}$	Concentrated near continuum ($\propto -1/n^2$).

Module 5

Macrostate, Microstate, Phase Space, and Ensembles

1. Macrostate, Microstate, Phase Space, and Ensembles

1.1 Basic Concepts of Statistical Mechanics

Statistical mechanics connects the microscopic behavior of individual atoms/molecules with the macroscopic thermodynamic properties (P, V, T, S) of a system.

Definitions: Macrostate, Microstate, and Phase Space

- **Macrostate:** The macroscopic state of a system specified by overall thermodynamic variables (e.g. (N, V, E) or (N, V, T)).
- **Microstate:** A specific detailed configuration of the system specifying the exact position ($\vec{r}_i$) and momentum ($\vec{p}_i$) of every individual particle.
- **Phase Space:** A $6N$ -dimensional conceptual space formed by 3 position coordinates (x, y, z) and 3 momentum coordinates (p_x, p_y, p_z) for each of N particles. The elementary phase cell volume in quantum mechanics is h^3 .
- **Thermodynamic Probability (W):** The total number of microstates corresponding to a given macrostate.

Boltzmann Entropy Relation

$$S = k_B \ln W \quad (83)$$

where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, and W is the thermodynamic probability.

1.2 Statistical Ensembles

An **ensemble** is a large collection of independent, macroscopically identical systems existing in different microstates.

Three Main Gibbs Ensembles:

Ensemble	Fixed Parameters	Physical System Example
Microcanonical	Number N , Volume V , Energy E (Rigid, isolated, insulated walls).	Isolated system with zero heat exchange.
Canonical	Number N , Volume V , Temperature T (Diathermal walls in heat bath).	Closed system exchanging heat with a reservoir.
Grand Canonical	Chem. potential μ , Volume V , Temp. T (Permeable, diathermal walls).	Open system exchanging both energy and particles.

Solved Problem 1: Microstates and Entropy

Problem: Calculate the number of microstates and thermodynamic probability W for distributing $N = 4$ distinguishable particles in 2 equal compartments. Find the entropy of the most probable state.

Solution:

- Total number of microstates $= 2^N = 2^4 = 16$.
- Macrostates are given by (n_1, n_2) where $n_1 + n_2 = 4$:

- $(4, 0) : W = \frac{4!}{4!0!} = 1$
- $(3, 1) : W = \frac{4!}{3!1!} = 4$
- $(2, 2) : W = \frac{4!}{2!2!} = 6$ (Most Probable Macrostate)
- $(1, 3) : W = 4$
- $(0, 4) : W = 1$

- Entropy of most probable macrostate $(2, 2)$:

$$S = k_B \ln(6) = (1.38 \times 10^{-23}) \times 1.7917 = 2.47 \times 10^{-23} \text{ J/K} \quad (84)$$

2. Density of States

2.1 Derivation of Density of States in 3D

The **Density of States** ($g(E)dE$) represents the number of available quantum states per unit volume in the energy range between E and $E + dE$.

3D Free Particle Density of States Derivation

In momentum space, momentum $p = \sqrt{2mE}$. The volume of momentum sphere of radius p is $V_p = \frac{4}{3}\pi p^3 = \frac{4}{3}\pi(2mE)^{3/2}$. The number of spatial-momentum phase cells of volume h^3 inside spatial volume V is:

$$N(E) = \frac{V \cdot \frac{4}{3}\pi(2mE)^{3/2}}{h^3} = \frac{4\pi V(2m)^{3/2}}{3h^3} E^{3/2} \quad (85)$$

Taking the derivative with respect to E gives $g(E) = \frac{dN(E)}{dE}$:

$$g(E)dE = \frac{4\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (86)$$

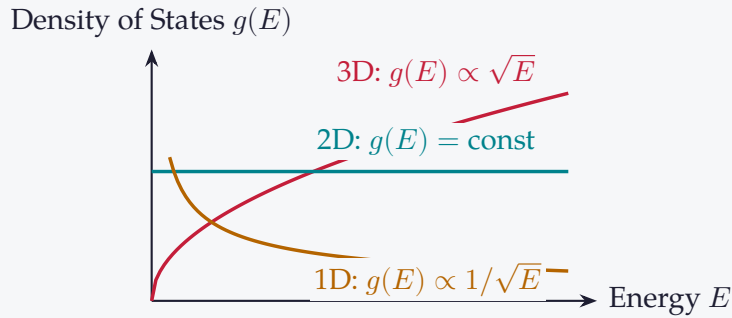
Including electron spin degeneracy $g_s = 2s + 1 = 2$ for electrons:

$$g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE \quad (87)$$

Density of States in Lower Dimensions:

1. **1D System:** $g_{1D}(E) \propto E^{-1/2}$.
2. **2D System:** $g_{2D}(E) = \text{constant}$ (independent of energy E).
3. **3D System:** $g_{3D}(E) \propto E^{1/2}$.

Density of States $g(E)$ in 1D, 2D, and 3D Systems



3. Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac Statistics

3.1 Qualitative and Quantitative Treatment

Depending on particle distinguishability, quantum spin, and wave function symmetry:

Three Distribution Functions

- **Maxwell-Boltzmann (MB):** Classical, distinguishable particles, no spin restriction:

$$f_{\text{MB}}(E) = A e^{-E/k_B T} \quad (88)$$

- **Bose-Einstein (BE):** Quantum, indistinguishable particles with integer spin ($s = 0, 1, 2, \dots$, Bosons):

$$f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1} \quad (89)$$

- **Fermi-Dirac (FD):** Quantum, indistinguishable particles with half-integer spin ($s = 1/2, 3/2, \dots$, Fermions), obeys Pauli Exclusion Principle:

$$f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1} \quad (90)$$

Comprehensive 8-Criterion Comparison Table:

Feature	Maxwell-Boltzmann	Bose-Einstein	Fermi-Dirac
Particle Nature	Classical, Distinguishable.	Quantum, Indistinguishable.	Quantum, Indistinguishable.
Spin	Any spin (Spin ignored).	Integral spin $(0, 1\hbar, 2\hbar)$.	Half-integral spin $(\frac{1}{2}\hbar, \frac{3}{2}\hbar)$.
Pauli Principle	Does not apply.	Does not apply.	Strictly obeys Pauli principle.
Occupancy/Cell	Unlimited $(0, 1, 2, \dots, \infty)$.	Unlimited $(0, 1, 2, \dots, \infty)$.	At most one particle (0 or 1).
Wave Function	No wave function overlap.	Symmetric under particle exchange.	Anti-symmetric under particle exchange.
Ground State	All particles rest at $E = 0$ as $T \rightarrow 0$.	Bose-Einstein Condensation at $T < T_c$.	Occupies up to Fermi level E_F at $T = 0$ K.
Physical Examples	Ideal gas molecules (He, Ar).	Photons, Phonons, α -particles, ^4He .	Free electrons, Protons, Neutrons, ^3He .

4. Fermi Energy and Bose-Einstein Condensation

4.1 Fermi Energy and Fermi-Dirac Function at $T = 0$ K

The **Fermi Energy** (E_F) is the highest filled energy level in a fermion system at absolute zero temperature ($T = 0$ K).

At $T = 0$ K:

$$f_{\text{FD}}(E) = \begin{cases} 1 & \text{if } E < E_F \\ 0 & \text{if } E > E_F \end{cases} \quad (91)$$

$$\text{Total number of free electrons } N = \int_0^{E_{F0}} g_e(E) dE = \int_0^{E_{F0}} \frac{8\pi V (2m)^{3/2}}{h^3} E^{1/2} dE = \frac{16\pi V (2m)^{3/2}}{3h^3} E_{F0}^{3/2}.$$

Fermi Energy and Average Energy at $T = 0$ K

Solving for E_{F0} in terms of electron density $n = N/V$:

$$E_{F0} = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3} \quad (92)$$

The total kinetic energy of N electrons at $T = 0$ K is $E_{\text{total}} = \int_0^{E_{F0}} E g(E) dE = \frac{3}{5} N E_{F0}$.
Average energy per electron at $T = 0$ K:

$$\langle E \rangle = \frac{3}{5} E_{F0} \quad (93)$$

Solved Problem 2: Fermi Energy of Copper

Problem: Copper has free electron density $n = 8.49 \times 10^{28} \text{ m}^{-3}$. Calculate: (a) Fermi energy E_F in eV, (b) Fermi velocity v_F .

Solution:

1. Fermi energy $E_{F0} = \frac{\hbar^2}{2m}(3\pi^2n)^{2/3}$: $3\pi^2n = 3\pi^2(8.49 \times 10^{28}) = 2.514 \times 10^{29} \text{ m}^{-3}$.
 $(3\pi^2n)^{2/3} = 3.983 \times 10^{19} \text{ m}^{-2}$.

$$E_{F0} = \frac{(1.054 \times 10^{-34})^2}{2 \times (9.11 \times 10^{-31})} \times 3.983 \times 10^{19} = 2.428 \times 10^{-18} \text{ J} = \mathbf{7.04 \text{ eV}} \quad (94)$$

2. Fermi velocity $v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2 \times 2.428 \times 10^{-18}}{9.11 \times 10^{-31}}} = \mathbf{1.57 \times 10^6 \text{ m/s}}$.

4.2 Bose-Einstein Condensation (BEC)

At extremely low temperatures below a critical transition temperature T_c , a substantial fraction of bosons collapse into the lowest quantum ground state ($E = 0$), forming a macroscopic quantum state known as **Bose-Einstein Condensate**.

Critical Temperature and Ground State Fraction

Critical BEC transition temperature T_c :

$$T_c = \frac{h^2}{2\pi m k_B} \left(\frac{N}{2.612 V} \right)^{2/3} \quad (95)$$

Fraction of bosons in ground state for $T < T_c$:

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2} \quad (96)$$

Quick Revision — Module 5

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Key Formulas and Metrics at a Glance

Boltzmann Entropy Relation	$S = k_B \ln W$ (W = thermodynamic probability)
Elementary Phase Cell	$d\Omega = h^3$ (quantum phase space volume)
3D Density of States	$g_e(E)dE = \frac{8\pi V(2m)^{3/2}}{h^3} E^{1/2} dE$
MB Distribution Function	$f_{\text{MB}}(E) = Ae^{-E/k_B T}$
BE Distribution Function	$f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$ (Bosons, Integer spin)
FD Distribution Function	$f_{\text{FD}}(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$ (Fermions, Half-spin)
Fermi Energy ($T = 0$ K)	$E_{F0} = \frac{h^2}{8m} \left(\frac{3N}{\pi V}\right)^{2/3} = \frac{h^2}{2m} (3\pi^2 n)^{2/3}$
Average Fermi Energy	$\langle E \rangle = \frac{3}{5} E_{F0}$ (at $T = 0$ K)
BEC Transition Temp.	$T_c = \frac{h^2}{2\pi m k_B} \left(\frac{N}{2.612 V}\right)^{2/3}$
BEC Condensate Fraction	$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2}$ (for $T < T_c$)

One-Line Definitions

- **Microstate:** A specific detailed microscopic specification of position and momentum coordinates for all particles in a system.
- **Ensemble:** A large virtual collection of independent, macroscopically identical systems in different microstates.
- **Fermi Energy (E_F):** The maximum energy state occupied by free electrons at absolute zero temperature ($T = 0$ K).
- **Boson:** A particle with integral quantum spin ($0, 1\hbar, 2\hbar$) obeying Bose-Einstein statistics and symmetric under particle exchange.
- **Bose-Einstein Condensation:** A phase transition where a macroscopic number of bosons collapse into the single lowest energy ground state at $T < T_c$.

Frequently Asked / Viva Questions

Q1. What is the difference between a macrostate and a microstate?

A macrostate specifies overall macroscopic variables (N, V, E); a microstate specifies exact microscopic position and momentum coordinates ($\vec{r}_i, \vec{p}_i$) of every particle.

Q2. What is phase space and what is its minimum volume in quantum mechanics?

Frequently Asked / Viva Questions (contd.)

Phase space is a $6N$ -dimensional space combining 3 position and 3 momentum axes per particle. Its minimum volume cell in quantum mechanics is h^3 .

Q3. Distinguish between Microcanonical, Canonical, and Grand Canonical ensembles.

Microcanonical keeps (N, V, E) fixed; Canonical keeps (N, V, T) fixed (heat exchange allowed); Grand Canonical keeps (μ, V, T) fixed (heat and particle exchange allowed).

Q4. State the basic difference between Bosons and Fermions.

Bosons have integral spin $(0, 1\hbar)$, symmetric wavefunctions, and do not obey Pauli principle. Fermions have half-integral spin $(\frac{1}{2}\hbar)$, anti-symmetric wavefunctions, and strictly obey Pauli principle.

Q5. Why does density of states in 3D vary as $\sqrt{E}$?

Because the momentum sphere volume $V_p \propto p^3 \propto E^{3/2}$. Taking the derivative $g(E) = dN/dE$ gives $g(E) \propto E^{1/2} = \sqrt{E}$.

Q6. Draw Fermi-Dirac distribution curve at $T = 0$ K and $T > 0$ K.

At $T = 0$ K, $f(E)$ is a step function ($f = 1$ for $E < E_F$, $f = 0$ for $E > E_F$). At $T > 0$ K, thermal agitation creates a symmetric tail around E_F where $f(E_F) = 1/2$.

Q7. What is the physical meaning of Fermi energy?

Fermi energy represents the highest energy level occupied by free conduction electrons in a metal at absolute zero temperature ($T = 0$ K).

Q8. Calculate the average energy of an electron gas at $T = 0$ K in terms of E_F .

$$\langle E \rangle = \frac{\int_0^{E_F} E \cdot E^{1/2} dE}{\int_0^{E_F} E^{1/2} dE} = \frac{\frac{2}{5} E_F^{5/2}}{\frac{2}{3} E_F^{3/2}} = \frac{3}{5} E_F.$$

Q9. Under what conditions do BE and FD statistics reduce to MB statistics?

At high temperatures ($T \rightarrow \infty$) and low particle densities ($n \rightarrow 0$), when $e^{(E-\mu)/k_B T} \gg 1$, both BE and FD distributions reduce to classical MB distribution $f(E) \approx e^{-(E-\mu)/k_B T}$.

Q10. What is Bose-Einstein Condensation (BEC)?

BEC is a low-temperature phenomenon where a macroscopic fraction of bosons condense into the single zero-momentum ground state at $T < T_c$.

Q11. Why can photons undergo BEC without fixing total particle number N ?

Photons have chemical potential $\mu = 0$ because photon number is not conserved (photons are freely created/absorbed by cavity walls).

Q12. Why do electrons in a metal not contribute significantly to heat capacity at room temperature?

Because Pauli exclusion principle restricts thermal excitation ($k_B T$) to only a small fraction ($\sim k_B T / E_F \approx 1\%$) of electrons near the Fermi level E_F .

Summary Comparison of Quantum Statistics

Property	Maxwell-Boltzmann	Bose-Einstein	Fermi-Dirac
Distribution Eq.	$f(E) \propto e^{-E/k_B T}$	$f(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}$	$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$
Ground State Occupancy	Non-zero classical energy.	Condensate (N_0 particles at $E = 0$).	Filled Fermi sphere up to E_F .
Wave Symmetry	Distinguishable (No symmetry).	Symmetric under exchange.	Anti-symmetric under exchange.